

Design Before You Build — Paper C

HARD • 45 MINUTES • 30 MARKS

NAME

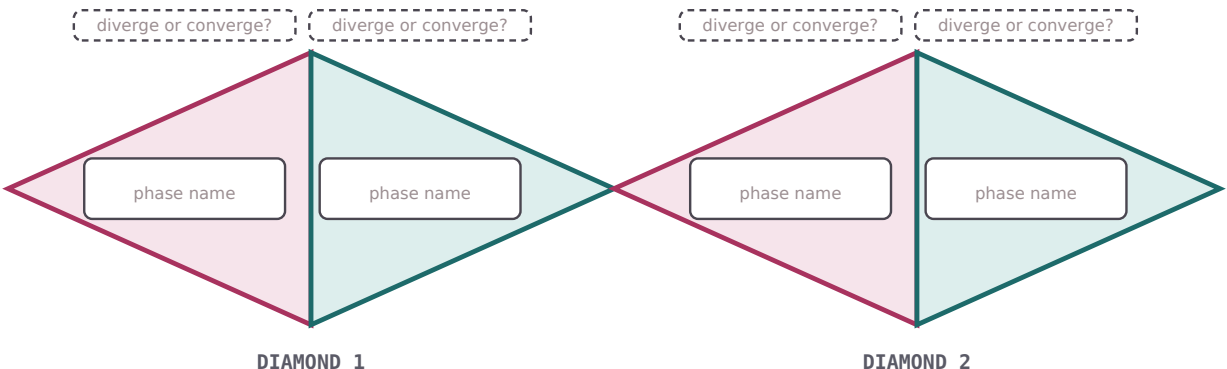
DATE

SCORE / 30

- Answer **every** question in the space provided. Show your working: method marks are awarded even when the final answer is wrong.
- No calculator and no computer. Every number on this paper works by hand.
- Where a question says **sketch**, draw plain boxes and labels only. Colour, real pictures and decoration earn nothing on a wireframe.
- Write **px** on every screen measurement. Give a contrast ratio as a number.
- The mark for each question is shown in square brackets on the right.
- No formula and no pass mark is given on this paper. If you cannot remember them, write down what you do know and work from there. Relative luminance L is printed wherever you need it.
- Several questions describe products and situations you may not have seen before. Apply the ideas you already have.

SECTION 1 – KNOW THE IDEAS

- 1.** A school canteen team is following the double diamond to design a pre-order app. The diagram is blank. After launch, a survey shows that students skip lunch because the break is too short, a cause nobody had researched.



- (a) Write the four phases in the boxes, in order. [2]
- (b) In each dashed box, write **diverge** or **converge**. [1]
- (c) Draw an arrow on the diagram from the end of the process back to the phase the team should return to, and name that phase. [1]

2. A temple trust wants to reduce crowding at its free-meal counter. After watching the counter for a week, the team writes thirty different ideas on sticky notes (a token system, two counters, a one-way queue...) and announces: *"Define is done."*
- (a) Name the phase the team is really in, and say whether it is diverging or converging. [1]
-
- (b) Explain what Define should have ended with, and why the team needs it before listing ideas. [2]
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3. For each fix to the canteen order form, write **UI** or **UX** and give a reason of no more than ten words.
- (a) Add "Step 2 of 4" and a progress bar to the form. [1]
-
- (b) Change the form text from 12 px to 16 px. [1]
-
- (c) Let a student repeat last week's lunch order with one tap. [1]
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SECTION 2 – WORK IT OUT

4. Three colour samples for the canteen app. L is relative luminance; the text colour and the background colour are printed with their L values.

Sample i



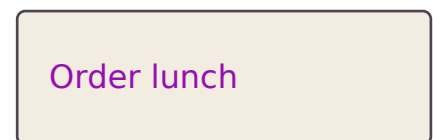
text #81B7F3 L = 0.45
back #153696 L = 0.05

Sample ii



text #153696 L = 0.05
back #D28151 L = 0.30

Sample iii



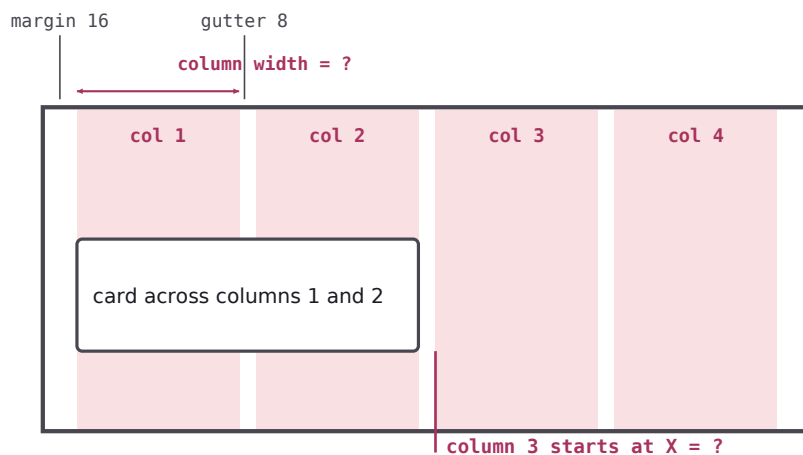
text #960FB1 L = 0.10
back #F3EDE1 L = 0.85

(a) Calculate the contrast ratio for each of the three samples. [3]

(b) State which sample fails for 16 px body text but would be allowed for a 32 px heading. [1]

(c) A student says: "Sample ii looks perfectly clear on my phone, so it passes." Give a one-sentence reply. [1]

5. A designer sets up a **4-column** layout grid on a phone frame **360 px** wide, with a margin of **16 px** at each side and a gutter of **8 px** between neighbouring columns. Figma shares the space that is left equally between the four columns.



FIGMA PANEL

Frame: W 360
Layout grid: 4 columns
Margin: 16 each side
Gutter: 8 between columns

Figma divides the space that is left equally between the four columns.

(a) Calculate the width of one column. [2]

(b) A card spans columns 1 and 2. Calculate its width. [1]

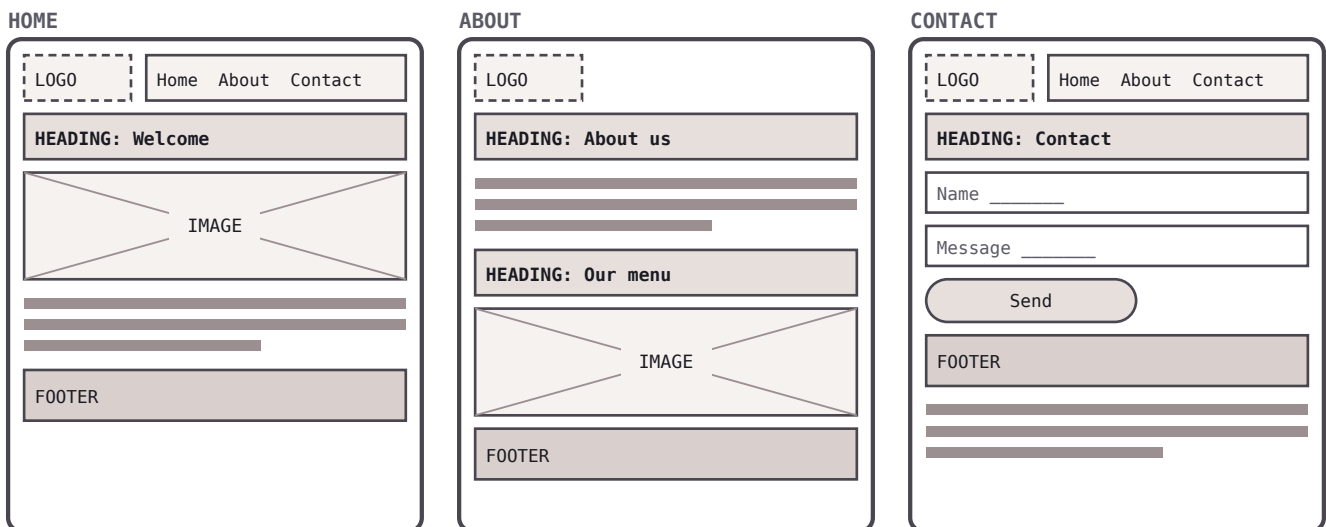
(c) Calculate the X position where column 3 starts.

[2]

SECTION 3 – JUDGE THE DESIGN

6. Tanvi draws first wireframes for the canteen’s website. Her brief: the same navigation at the top of every page; exactly one heading per page, before the content; a form on Contact only; a button on Home after its heading; every page ends with its footer. Her drawings contain **four** mistakes against that brief. For each, name the page and say which part of the brief is broken.

[4]



7. Before the first meeting, Farhan spends a week turning his wireframe into a finished-looking design with the canteen’s orange, real photographs of dosa and a custom font. The principal spends the whole meeting arguing about the shade of orange and never notices that the Home page has no way to reach the menu.

(a) Name the mistake in Farhan’s approach.

[1]

(b) Explain how it caused what happened in the meeting.

[1]

(c) State what he should have shown instead.

[1]

8. A school runs a bus for **36 families**. In a survey, **27** parents said they wait at the gate for the bus, that the wait averages **18 minutes**, and that the bus is anything from **10 to 25 minutes** late. Write **one sentence**, a problem statement for the Define phase. It must say who is affected, what goes wrong and why it matters, use one figure from the research, and suggest no solution. [3]

End of paper. Check that every screen measurement carries px, and that every UI or UX answer says why, not just which.